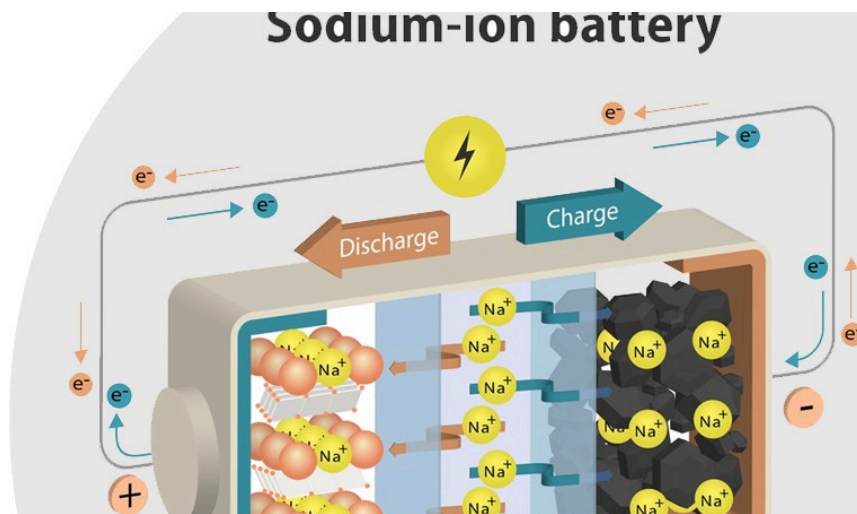


ARE SODIUM-ION BATTERIES THE FUTURE FOR GRIDS, HOMES, AND COM- PACT EVs?



Are Sodium-Ion Batteries the Future for Grids, Homes, and Compact EVs?



Neeraj Kumar Singal

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Na-ion (sodium-ion) batteries are one of the most affordable and widely available substitutes for Lithium-ion (Li-ion) batteries. Although, there is sufficient lithium in the world to meet the goals for electrification, there is an urgent need for an alternative due to tightening supply chain constraints and rising demand. A Na-ion battery cell should cost between USD 40 and 80 cents per kWh, while a Li-ion battery cell should cost an average of USD120 per kWh. Na-ion batteries have a longer cycle life, faster charging times, and a safer operating temperature range. They are heavier and bulkier due to their lower energy density. However, it is still sufficient for city-range EVs at 160 Wh/kg (which should increase), and Chinese manufacturers have already revealed Na-ion compact EVs with a 250 km range. Production capacity is expected to increase from 42 GWh/year in 2023 to 186 GWh/year by 2030, which is sufficient to power the 4.6 million EVs produced annually. Furthermore, size is irrelevant for home storage and stationary grid. This is a tale about Na-ion, but it's also about how crucial it is for the world to innovate energy solutions in place of conventional ones, guaranteeing a diverse supply chain and reasonable costs. The global energy transition towards net-zero will necessitate adjustments in energy production and consumption, with a focus on renewables. The eventual direct and indirect electrification of energy end-use sectors (buildings, transportation, and industry) will be one of these changes.

Unlocking Success: The Key to a Seamless Transition is Energy Storage

One cannot stress how crucial storage is to a smooth transition. According to IRENA's 1.5°C Scenario, battery storage will be necessary to provide the power system with a great deal of flexibility. By 2030, it will reach nearly 360 GW, and by 2050, 4,100 GW. Beyond the power industry,

however, battery storage will be essential to the decarbonization of end-use industries. Take electric vehicles, which are expected to make up 90% of all road transportation by 2050.